Security: protecting money, data and trust

Four risks we found, and what Nolifa does about each one

1 Delivery PIN

Risk: unlimited guesses at a 4-digit PIN.

Fix: 3 tries. The third wrong try locks the order, opens a dispute and freezes the money until a mediator rules.

BUILT

2 Escrow

Risk: the app decides when money moves, so a modified app could release it.

Fix: a licensed provider (FNB) holds the funds. A server checks the PIN and is the only thing that can release.

DESIGNED

3 Data on the phone

Risk: IDs, health records and orders sit on the device. Under POPIA they are special personal information.

Fix: AES-256-GCM encryption. The key is kept in the Keychain or Keystore, away from the data.

BUILT

4 One voice each

Risk: five accounts can vote five times or confirm a fake outbreak.

Fix: only verified people vote on water or confirm an outbreak. One verified number per account.

BUILT, BACKEND NEEDED

Also in place: salted password hashes, the PIN stored only as a hash, report locations rounded to about 5 km, proof photos at pickup and delivery.

Honest limit: payments, escrow and identity checks are simulated in this prototype. Moving them to a licensed provider is the first production step.